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An Experimental Assessment of Treatments for Cyclical Data

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Abstract

Cyclical features like hour-of-day are widely used as predictor variables in machine learning. It is often recommended that cyclical features be encoded using a sine/cosine method. This encoding has the appealing property that two similar hours will be nearby in feature space. Here we experimentally assess the performance of sine/cosine encoding and other encodings for cyclical features. We consider both real-world and synthetic data sets, and both classification and regression algorithms. Our main result is that the power of an encoding depends heavily on the particular machine learning algorithm being used. We also define a new similarity encoding method works well when decision trees are used.

1. Introduction

In applying machine learning to data sets with a time component, features like "hour of day" and "day of week" are commonly used. For example, one might want to predict whether a car accident will involve substance abuse based on the hour of day and day of the week. Figure 1 shows some relevant data [1] from Montgomery County, Maryland. In a set of training data for use in machine learning, the value of "hour of day" can be encoded as an integer ranging from 0 to 23. A potential problem in this representation is that while midnight and 11 pm are one hour apart, the difference between their encodings (0 and 23) is large.

One web source [2] refers to this encoding of the hour of the day as a common beginner mistake. A recommended alternative encoding (for example, in [2, 3]) is the "sin/cos encoding", in which the integer hour-of-day value is replaced by two variables. If h is the simple integer encoding of hour-of-day, then the corresponding value in the sin/cos encoding is the pair

$$sin\left(\frac{2\pi h}{24}\right), cos\left(\frac{2\pi h}{24}\right)$$

For example, the encoding of the hour $h = 10$ would be $(0.5, -0.866)$. The term $2\pi h/24$ serves to transform the hour of day value to units of radians. Fig. 2 shows a plot of

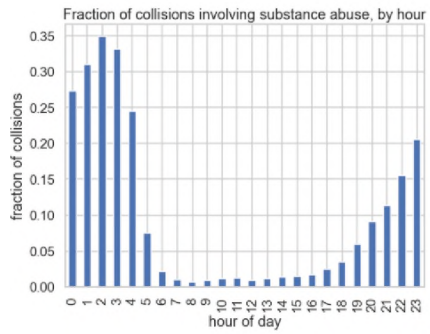

Figure 1. Vehicle collision data from Montgomery County, Maryland.

the hour-of-day values 0 through 23 in this encoding, with midnight being the point at the top. Clearly, the Euclidean distance between the points for 11 pm and midnight is the same as the distance between the points for midnight and 1 am.

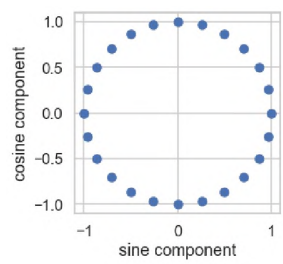

Figure 2. The sin/cos encoding of hour-of-day values 0 through 23.

The sin/cos encoding of "cyclical features" like time of day is appealing, but does it actually help improve the predictive power of machine learning algorithms? Furthermore, are the benefits of the sin/cos encoding dependent on the type of machine learning algorithm being used? Would a logistic regression classifier benefit more from the sin/cos encoding than, say, a classification tree? These are important questions for practitioners of machine learning. In [3] it is stated that the sin/cos encoding is not effec-

tive for decision-tree based algorithms because these algorithms process only a single variable at a time.

Unfortunately, nothing in the research literature seems to answer these questions. Some work does look at narrower questions, however. For example, Flury and Levri [4] look at the sin/cos encoding for logistic regression in application to topics in ecology. However, no comparison is made to alternative encodings or other machine learning algorithms. Moon *et al* [5] use sin/cos encoding as part of a power forecasting problem but do so without discussion or citations.

Another question is whether other encodings exist that are effective for machine learning. For example, is one-hot encoding effective? With one-hot encoding, a single hour-or-day feature becomes 24 new variables, each with value 0 or 1. In this encoding, any notion of the distance between two hours of the day is lost.

In this paper, we describe our work in testing alternative encodings of cyclical features in machine learning. We consider both real-world and synthetic data sets, and classification and regression problems. Besides the "raw" encoding, sin/cos encoding, and one-hot encoding methods, we also consider an encoding based on the concept of similarity feature, and describe a new method for selecting a "landmark" when using this encoding.

2. Data sets

We use the following data sets in our experiments:

- Occupancy data [6]. This data set contains variables related to the conditions of an office, such as temperature and humidity, and also contains a variable indicating whether the office is occupied.
- Collision data [1]. This data set contains information about collisions in Montgomery County, Maryland. Example variables include the time of the collision, the type of collision, and the injury severity.
- Air quality data [7]. This data set concerns air quality in London. Example variables include the time and the concentration of various pollutants.
- Crime data [8]. The data set contains San Francisco Police Department incident reports for the year 2019. Example variables include time of the report and incident category.
- Appliance data [9]. The data set consists of 4.5 months of measurements from wireless sensors in a household. Example variables include time, temperature, humidity, and energy consumption from appliances and lights.

We derive training data sets from each of these data sets by selecting predictor and target variables from among the available columns. (We refer to a variable, like hour-of-day, used as data in making predictions as a *predictor variable*. We refer to the variable being predicted as a *target variable*.) For example, from the occupancy data we can create a training data set for classification by using "hour of day" as the predictor variable and "occupancy" as the target variable. Or, we can create a training set for regression by using "hour of day" and "humidity" as predictor variables and "light" as the target variable.

For classification we also created several *synthetic* data sets. To do so we defined the conditional probability of the target variable having value 1 given an hour of the day. We then sampled many predictor values from a uniform distribution, and sampled a target value for each predictor value from a Bernoulli distribution for the hour.

In all we used 10 classification data sets: 4 real-world data sets with a single predictor, 3 real-world data sets with multiple predictors, and 3 synthetic data sets with a single predictor. We also used 10 regression data sets: 7 real-world data sets with a single predictor and 7 real-world data sets with multiple predictors. In every data set there was one predictor for the hour-of-day.

To preprocess the training data for classification, we remove rows with missing data, "balance" the data set using under-sampling so that there are equal numbers of rows with target value 0 and target value 1, and scale the predictor values. The same process is used for regression, except that instead of balancing the data, the target variable is scaled to allow comparison of mean squared error values.

3. Encoding cyclical data

In our tests we compare four encodings of cyclical data. Fig. 3 shows the encodings of some hour-of-day values.

ordinal	sin/cos	one-hot	similarity
0	0.0,1.0	1,0,0,0,0,...,0	2
1	0.26, 0.97	0,1,0,0,0,...,0	1
2	0.5, 0.87	0,0,1,0,0,...,0	0
3	0.707, 0.707	0,0,0,1,0,...,0	1
...	...	...	...

Figure 3. Encodings of some hour-of-day values. A landmark value of 2 am is used for the similarity encoding.

The first is *ordinal encoding*. In this encoding, we simply represent the hour of day as an integer from 0 to 23.

Second is *sin/cos encoding* as described in Section 1. The hour of day is represented as two float values.

Third is *one-hot encoding*, also known as the dummy variable or indicator variable method. The hour of day is represented as 24 binary values; one for each hour of the day. (Actually, only 23 values are needed, as the 24th can be derived from the other 23, since exactly one will be 1.)

Fourth is *similarity encoding*. In this encoding the hour of day is a single value representing the distance to a specific time, which is called the *landmark*. If we use 4 pm

as the landmark, and let the distance between two hours be the absolute value of their difference, then the encoding of 3 pm would be 1, the encoding of 4 pm would be zero, and the encoding of 5 pm would be 1.

When similarity encoding is used a landmark value must be identified. Selecting a landmark manually is tedious and unreliable. We have developed a heuristic method for landmark selection that often works well. The idea is to compute the "center of mass" of the data, where the target values represent mass and the predictor values represent location.

To apply this idea to a cyclical variable, we use the following method:
1. Weight each sin/cos-encoded hour by its mean target value.
2. Compute the center of mass in this 2D space
3. Transform the center of mass back to an hour in 1D space.

For example, suppose we have a classification data set in which the average target value for hour 23 (11 pm) is 1/3 and the average target value for hour 1 (1 am) is 2/3 (and no other hours are present in the data). Mapping hours 23 and 1 to 2D space with the sin/cos encoding we obtain points (-0.26, 0.97) and (0.26, 0.97). Using the weights 1/3 and 2/3, their weighted average is (0.086, 0.97). Mapping this point back to its ordinal encoding we obtain the landmark: hour 0.34 (about 20 minutes after midnight).

The distance between an hour of day and the landmark is computed taking the cyclic nature of the data into account. In other words, the distance from hour 23 to landmark hour 1 is 2, not 22.

A benefit of the similarity encoding compared to the sin/cos encoding is that only one value is needed for the similarity encoding, not two. Also, the values produced by the similarity encoding are easier to interpret. For example, it is easy to interpret a predictor that represents the number of hours before or after 10 pm.

4. Machine learning algorithms

A property of the sin/cos encoding is that the distance between the encoding of 11 pm and midnight is the same as the distance between the encoding of midnight and 1 am. Is this property important in machine learning, and, if so, for which machine learning algorithms? In this section we review the machine learning algorithms we use in our experiments, focusing on the underlying model used to make predictions. We assume basic familiarity with machine learning.

In our experiment we use machine learning algorithms that are simple, widely used, and we include both parametric and non-parametric algorithms. For classification, we use logistic regression, k-nearest-neighbor classification, classification trees, and support-vector classifiers. Let us compare how each algorithm works in the case of hour-of-day encoded with alternative encodings. We focus on the ordinal and sin/cos encodings.

Logistic regression. In logistic regression, a prediction is the logistic function applied to a linear combination of the predictor variables. (The output ranges from 0 to 1, and can be converted to either 0 or 1 by using a threshold.) This means, with the ordinal encoding, the prediction for hour 23 will be as different from the prediction for hour 0 as is possible in the model. We expect the sin/cos encoding to work well with logistic regression.

k-nearest neighbors (KNN) classification. In KNN, the predicted value for an hour depends on the labels of "nearby" hours. Commonly, the distance is computed as the Euclidean distance, so with the ordinal embedding the absolute difference of the hours is used. Again, the sin/cos encoding is appealing with KNN. However, KNN does not work well with categorical variables. The problem is that there may be many nearest neighbors when categorical variables are present. How a KNN classifier behaves in this case depends on the implementation.

Classification trees. With a classification tree, the predicted value for an hour will depend on simple conditions related to the hour. If ordinal embedding is used, the conditions leading to a prediction might be $hour > 22$ and $hour < 2$. This suggests that classification trees are unlikely to benefit much from the sin/cos encoding compared to the ordinal encoding.

Support vector machine (SVM) classification. The model underlying a SVM is linear, but non-linear decision boundaries can be achieved through the used of a "kernel" function, through which a support vector machine can implicitly transform the predictor variables. The commonly-used radial basis function kernel implicitly transforms predictor variables using the same ideas as in our similarity encoding, but with many different landmarks. Therefore, we expect that SVM classifiers will not benefit much from sin/cos encoding.

For regression we use machine learning algorithms related to the four above: *Linear regression* is like logistic regression, but the logistic function is not applied. *KNN regression* is like KNN classification, but the predicted numeric value for an hour is the mean of the target values of nearby hours. A *regression tree* is like a classification tree, but each leaf node is labeled with a numeric value that is used as the prediction. *SVM regression* is like SVM classification, and supports the same kernel functions. We expect the usefulness of the sin/cos encoding for each regression algorithm to be similar as the usefulness for the algorithm's sibling classification algorithm.

5. Experimental setup

Our experimental set up is simple. For classification, we have a collection of data sets, a collection of classification algorithms, and a collection of encoders. For each combination we compute accuracy and F1 score for the algorithm using 10-fold cross validation. Here is pseudo-code for the process:

```
for every classification data set:
  for every encoder:
    encode the cyclical predictor variable
    for every classifier:
      scale the data
      compute accuracy and F1 score
      record the result of the test
```

The process for regression algorithms is almost identical, except that we measure mean squared error rather than accuracy and F1 score. The results of the experiments are stored in a table, with each row an experiment and each column a parameter, such as the encoder type. Overall, we ran just over 600 experiments.

In each data set, the only cyclical feature is hour of day. Two versions of the feature are used, one of type integer, and one of type float in which hour-of-day is treated as a continuous variable. The one-hot encoding of the cyclical predictor variable is not used when the hour of day is represented as a floating-point value.

6. Experimental results

Due to space we can only summarize our experimental results. Figures 4 and 5 show accuracy results for two data sets, and are representative of other results. (Plots can be magnified in the document reader for more detail.)

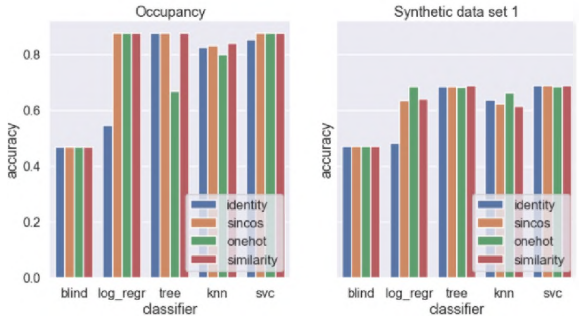

Figure 4. Classification: results for two data sets with only hour-of-day as the predictor variable. The real-world Occupancy data set (left) and a synthetic data set (right).

In case of logistic regression, one-hot encoding worked the best for both real and synthetic data sets when hour-of-day values are integers. Sin/cos encoding did as good as one-hot encoding on real-world data sets, but in the case

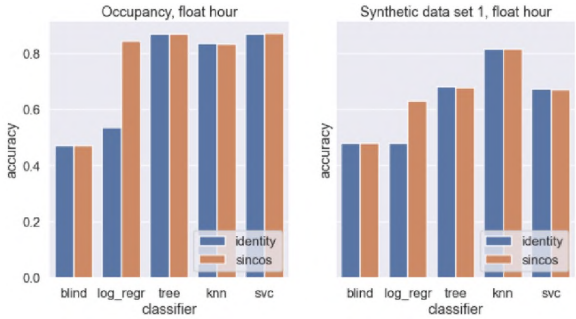

Figure 5. Classification: results for the same data sets as in Fig. 4, but with hour-of-day represented as a floating-point value.

of synthetic data, one-hot encoding was slightly better than sin/cos. When using floating-point hour-of-day values, in some cases sin/cos encoding performed better than ordinal encoding by a margin of 0.2, while in other cases both worked the same.

For classification trees we found that sin/cos and ordinal encoding both perform the same on real and synthetic data when working with integers. It is worth mentioning that the one-hot encoder performed poorly than the others. Similarity encoding was as good as sin/cos and ordinal embedding with synthetic data. When using floating-point hour-of-day values, ordinal and sin/cos encoding give about the same accuracy.

KNN classifier showed varying results with encoders depending on the data set used. For floating-point hour-of-day values, ordinal and sin/cos encoding gave similar results. For the support vector classifier, choice of encoder had little impact on accuracy. Ordinal and sin/cos encoding worked the same for floating-point hour values.

Figures 6 and 7 are representative results of our regression experiments.

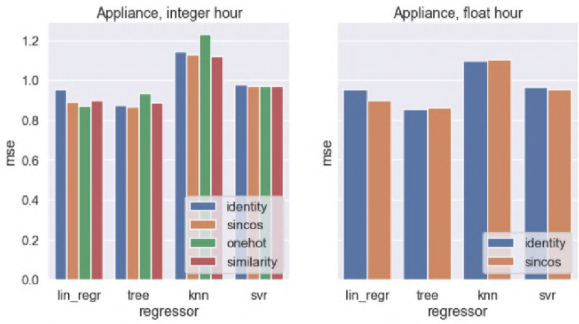

Figure 6. Regression: results for two data sets with only hour-of-day as the predictor variable. The real-world Appliance data set is shown, with both integer hour-of-day (left) and floating-point hour-of-day (right).

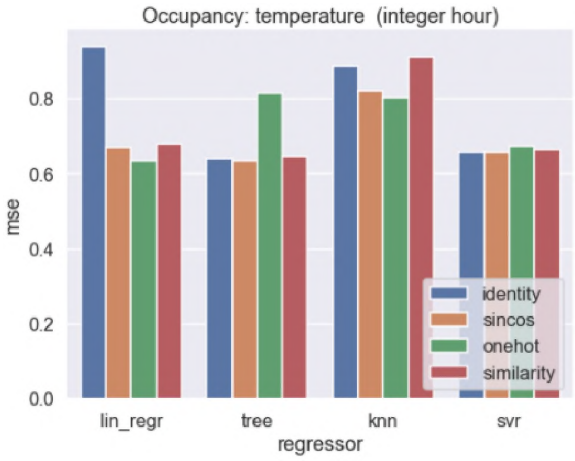

Figure 7. Regression: results for the real-world Occupancy data set.

With linear regression, ordinal encoding performed poorly. When hour-of-day was captured as integer value, one-hot encoding and similarity encoding worked best. When hour-of-day was captured as a floating-point value, sin/cos encoding worked best.

With regression trees and integer hour-of-day values, ordinal, sin/cos and similarity encoding performed similarly, while one-hot was worse. With floating-point hour-of-day values, sin/cos and ordinal encoding worked best.

Like its classification counterpart, KNN regressor showed that best encoding was dependent on the data set when using integer hour-of-day values. For floating-point values, we saw that ordinal and sin/cos encodings worked the same.

Lastly, the support vector regressor was not affected by the choice of encoding at all. All encodings resulted in the same mean square error.

7. Conclusions

Summarizing our findings, we found that logistic regression and linear regression benefit from using sin/cos encoding, and suffer when ordinal encoding is used. Classification and regression trees suffer from the choice of one-hot encoding. KNN regression and classification do not perform well with integer hour-of-day values. When hour-of-day values are floating-point, the raw encoding performs almost as well as sin/cos encoding.

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